DATA SCIENCE PROJECT REPORT

Customer Churn Prediction and Retention Strategy

A Machine Learning Solution for Business Analytics: XGBoost, SHAP Explainability, Profit-Based Decision Thresholds and an Interactive Streamlit Dashboard

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Dataset: IBM Telco Customer Churn (Kaggle)

Tools: Python, pandas, scikit-learn, XGBoost, SHAP, Matplotlib, Seaborn, Streamlit

Executive Summary

Customer churn is one of the most expensive problems in subscription businesses, because acquiring a new customer typically costs far more than keeping an existing one. This project builds an end-to-end analytics solution for a telecommunications company: it predicts which customers are likely to leave, explains why, and converts those predictions into a profit-optimized retention campaign.

Key results

- **Predictive performance:** XGBoost reached a test ROC AUC of 0.835 (5-fold cross-validated AUC of 0.846). Logistic Regression and Random Forest performed almost identically, so the model choice was driven by interpretability and tooling rather than raw accuracy.
- **Threshold tuning matters more than algorithm choice:** lowering the decision threshold from 0.50 to 0.37 raised recall on churners from 0.54 to 0.71 for XGBoost, and similarly for the other two models.
- **Main churn drivers:** short tenure, month-to-month contracts, fiber optic internet, electronic check payment and high monthly charges raise risk. Long contracts, online security, tech support and long tenure reduce it.
- **Business value:** under baseline assumptions (customer value 500, contact cost 50, campaign success 30%), the profit-optimal threshold of 0.37 yields an expected profit of 16,500 currency units on the 1,407-customer test set.
- **Robustness:** a sensitivity analysis shows that profit depends strongly on campaign success rate, customer value and contact cost, which identifies exactly which assumptions must be validated with a real A/B test before roll-out.
- **Deployment:** the final model is wrapped in an interactive Streamlit dashboard that returns a churn probability, the top SHAP reasons and recommended retention actions for any customer profile.

1. Introduction and Business Problem

The company in this case study sells phone and internet services on subscription. Roughly one in four customers leaves, and each departure removes recurring revenue. Retention campaigns (discounts, upgrades, outreach) can save some of these customers, but they cost money and are wasted on customers who would have stayed anyway.

The business question is therefore not simply "who will churn?", but "whom should we contact so that expected profit is maximized?". The project addresses this in four steps:

- Predict the probability that each customer will churn.
- Explain which factors drive that probability, so the business can act on root causes.
- Choose a contact threshold by maximizing expected campaign profit rather than statistical accuracy.
- Deliver the result as a tool that non-technical staff can use.

2. Data

The analysis uses the public IBM Telco Customer Churn dataset from Kaggle. Each row is one customer, described by demographics, subscribed services, account information and the target label "Churn" (Yes/No).

Item	Description
Raw records	7,043 customers, 21 columns
After cleaning	7,032 customers (11 rows removed because TotalCharges was blank)
Target	Churn (1 = left, 0 = stayed); churn rate 26.6%
Demographics	Gender, SeniorCitizen, Partner, Dependents
Services	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
Account	tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges

The target is imbalanced (73.4% stayed, 26.6% left). For this reason accuracy alone is misleading: a model that predicts "no churn" for everyone would already be about 73% accurate. Evaluation therefore focuses on ROC AUC, recall, precision and F1 for the churn class.

3. Methodology

3.1 Preprocessing

- Converted TotalCharges from text to numeric and dropped the 11 rows with missing values.
- Dropped the customerID column, which carries no predictive information.
- Encoded the target as 0/1 and one-hot encoded all categorical variables (drop-first).

3.2 Train/Test Split

The data was split 80/20 with stratification on the target, giving 5,625 training and 1,407 test customers with the same churn rate in both sets. A fixed random seed (42) ensures reproducibility.

3.3 Models

Three models were compared: Logistic Regression (with standard scaling), Random Forest (300 trees, max depth 8) and XGBoost (300 trees, learning rate 0.05, max depth 4). Each was evaluated with stratified 5-fold cross-validation on the training set and then on the held-out test set.

3.4 Evaluation

ROC AUC measures ranking quality independent of any threshold. Recall (the share of real churners found) and precision (the share of flagged customers who really churn) were reported at both the default 0.50

threshold and the tuned 0.37 threshold. Final decisions were evaluated in business terms through an expected-profit function (Section 7).

4. Exploratory Data Analysis

Exploratory analysis was done before modeling to understand the structure of the data and to form hypotheses that the model could later confirm or reject.

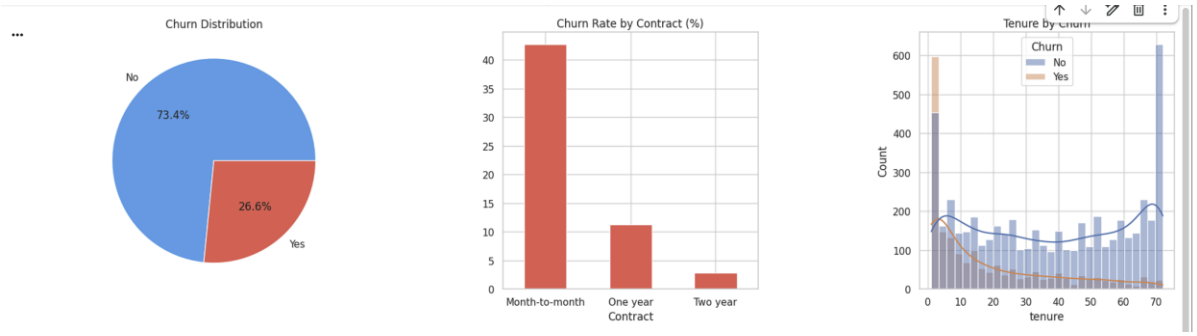


Figure 1. Churn distribution, churn rate by contract type, and tenure distribution by churn status.

Figure 1 shows three important patterns. First, 26.6% of customers churn. Second, contract type is the strongest single separator: the churn rate is roughly 42% for month-to-month contracts, about 11% for one-year contracts and about 3% for two-year contracts. Third, churners concentrate in the first months of tenure, while the large group at the far right of the tenure axis consists of loyal long-term customers.

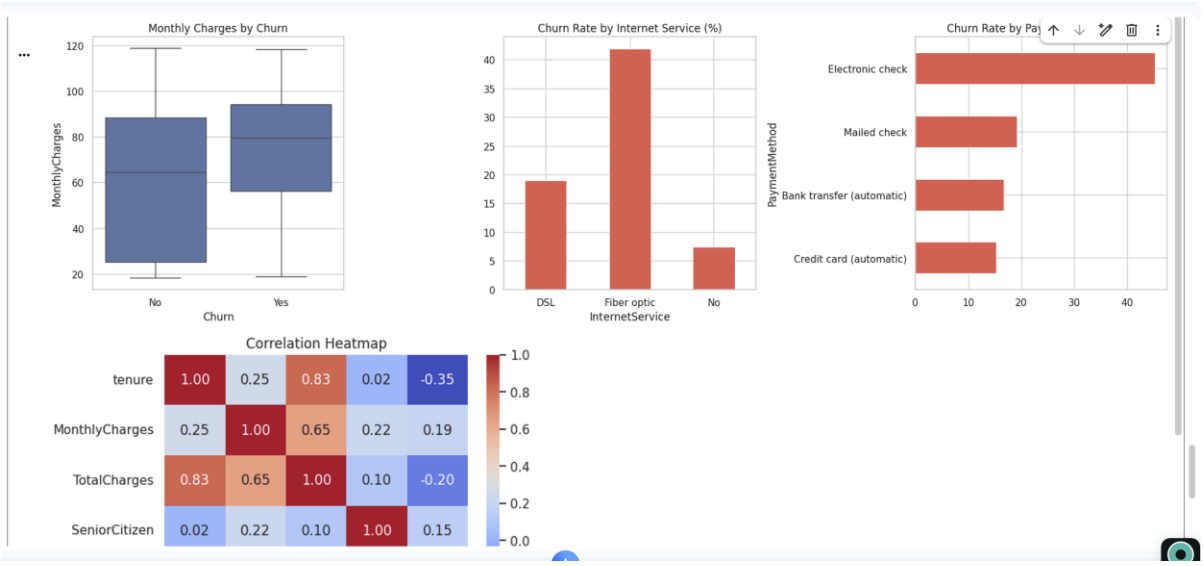


Figure 2. Monthly charges by churn, churn rate by internet service and payment method, and correlation heatmap of numeric features.

Figure 2 adds further evidence:

- **Price:** the median monthly charge of churners (about 80) is clearly above that of non-churners (about 65).

- **Internet service:** fiber optic customers churn at roughly 42%, against about 19% for DSL and about 7% for customers without internet. The premium product has the highest churn, which suggests a price or quality problem.
- **Payment method:** electronic check users churn at about 45%, while customers on mailed check, bank transfer or credit card churn at roughly 15% to 19%.
- **Correlations:** tenure is negatively correlated with churn (-0.35). Tenure and TotalCharges are strongly correlated (0.83) because TotalCharges is roughly tenure times monthly charges. This multicollinearity is harmless for tree-based models but would need attention in a plain Logistic Regression.

Note: percentages quoted from the charts are approximate.

5. Modeling Results

5.1 Baseline XGBoost at the Default Threshold

With the default 0.50 threshold, XGBoost achieved 79% accuracy and an AUC of 0.835. For the churn class, however, precision was 0.62 and recall only 0.54, meaning that almost half of the customers who actually left were never flagged. For a retention use case this is the costly kind of error.

Class	Precision	Recall	F1
Stayed (0)	0.84	0.88	0.86
Churned (1)	0.62	0.54	0.58

5.2 Effect of the Decision Threshold

Lowering the threshold flags more customers: recall rises and precision falls. The table below shows this trade-off for XGBoost on the test set.

Threshold	Precision	Recall	F1	Flagged
0.20	0.464	0.840	0.598	676
0.25	0.482	0.797	0.601	618
0.30	0.506	0.770	0.611	569
0.35	0.548	0.730	0.626	498
0.40	0.585	0.671	0.625	429
0.45	0.583	0.599	0.591	384
0.50	0.618	0.540	0.576	327
0.55	0.652	0.465	0.543	267
0.60	0.689	0.390	0.498	212

5.3 Model Comparison

Model	CV AUC	Test AUC	Recall @0.5	Prec. @0.5	Recall @0.37	Prec. @0.37	F1 @0.37
Logistic Regression	0.846	0.836	0.575	0.648	0.706	0.556	0.622
Random Forest	0.847	0.840	0.476	0.636	0.687	0.572	0.625
XGBoost	**0.846**	**0.835**	**0.540**	**0.618**	**0.714**	**0.567**	**0.632**

All three models perform almost identically: cross-validated AUC differs by only 0.001, and the test AUC spread (0.835 to 0.840) is within the noise expected from a test set of 1,407 customers. Two conclusions follow:

- The performance gain comes from threshold tuning, not from the choice of algorithm. At 0.37, recall for all three models is about 0.69 to 0.71, compared with 0.48 to 0.58 at 0.50.
- XGBoost was selected as the final model for its slightly higher F1 and recall at the tuned threshold and its native compatibility with SHAP, not because it is significantly more accurate. A simple Logistic Regression would be an equally defensible choice where maximum simplicity is required.

6. Model Interpretation with SHAP

Predictions alone are not actionable unless managers understand them. SHAP (SHapley Additive exPlanations) decomposes each prediction into the contribution of every feature. In the summary plots below, each dot is one customer; the horizontal position shows whether the feature pushed that customer's churn risk up (right) or down (left), and the color shows the feature value (red = high or "Yes", blue = low or "No"). Values are on the log-odds scale. Features are sorted by importance.

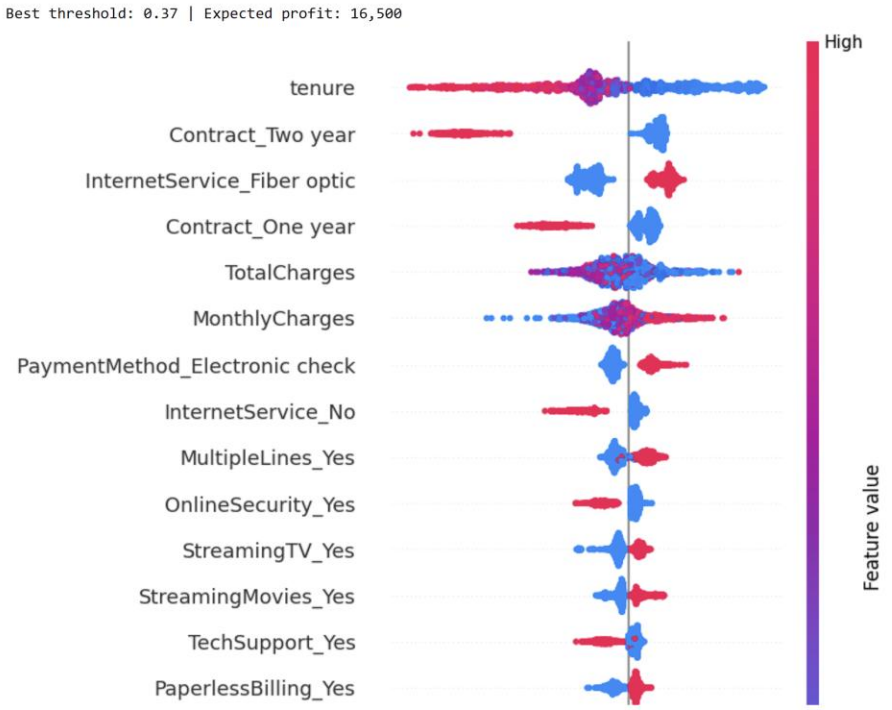


Figure 3. SHAP summary plot: most important features.

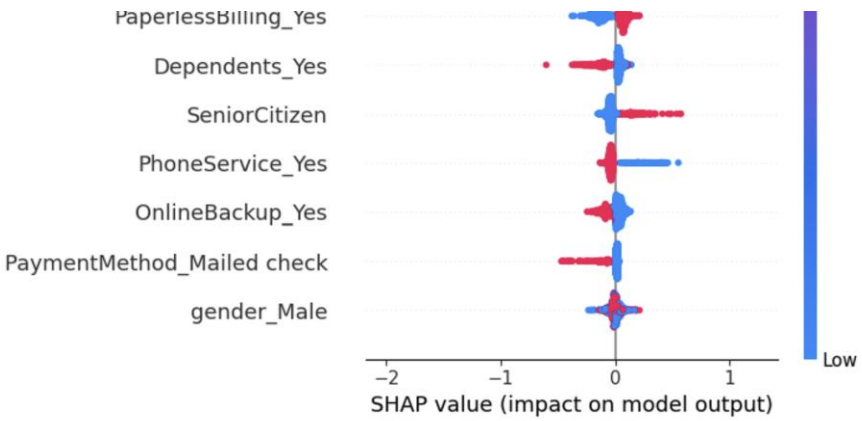


Figure 4. SHAP summary plot: less important features (continuation).

6.1 Factors that Increase Churn Risk

- **Short tenure:** the strongest driver. New customers are by far the most likely to leave.
- **Month-to-month contract:** having a one-year or two-year contract strongly reduces risk, so its absence raises it.
- **Fiber optic internet:** raises risk, consistent with the EDA.
- **Electronic check payment:** raises risk; mailed check lowers it.
- **High monthly charges:** tend to push risk up.
- **Senior citizens and paperless billing:** small positive effects on risk.

6.2 Factors that Reduce Churn Risk

- Long tenure and long-term (one- or two-year) contracts.
- Add-on services such as Online Security, Tech Support and Online Backup.
- Having dependents, and not subscribing to internet at all.

6.3 Fairness Observation

Gender has essentially no influence on predictions (its dots are centered on zero and intermixed), which is reassuring from a fairness standpoint.

6.4 Correlation Is Not Causation

SHAP explains what the model learned, not what causes churn. Long contracts, for example, may simply attract customers who were already loyal, and add-on services may signal more engaged customers. Recommendations derived from the model should therefore be validated experimentally before being treated as causal levers.

7. From Prediction to Profit: Business Threshold Optimization

7.1 Profit Model

For a given threshold, every flagged customer is contacted at a cost, and a fraction of the true churners among them is saved. Expected profit is:

Profit = (True positives × Success rate × Customer value) − (Flagged customers × Contact cost)

The baseline assumptions are illustrative and must be replaced with company data in practice:

Parameter	Baseline	Meaning
Customer value	500	Value of retaining one churning customer
Contact cost	50	Cost of one retention contact
Campaign success rate	30%	Share of contacted churners who are saved

A contact pays off roughly when the customer's churn probability exceeds cost ÷ (value × success), which is 50 ÷ (500 × 0.30) ≈ 0.33 in the baseline. Scanning all thresholds on the test set gave an optimum of 0.37 with an expected profit of 16,500, compared with a profit that is clearly lower at the default 0.50 threshold.

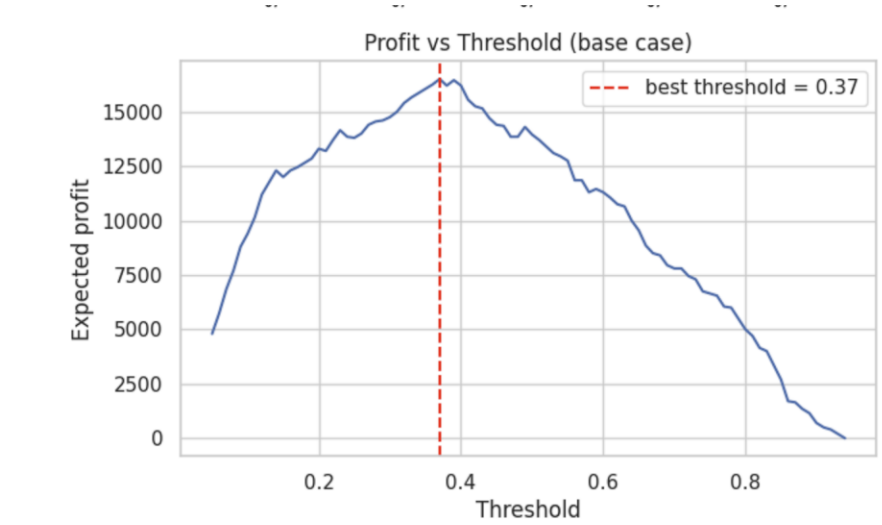


Figure 5. Expected profit versus decision threshold (baseline case).

The profit curve is bell-shaped with a broad peak: thresholds between roughly 0.30 and 0.50 still deliver about 13,000 to 16,500. The decision is therefore robust to small errors in the threshold, and the small wiggles in the curve are the natural noise of a 1,407-customer test set.

7.2 Sensitivity Analysis

Because the three business inputs are assumptions, each was varied while the others were held at baseline.

Campaign success rate

Success rate	Optimal threshold	Max profit
15%	0.76	1,150
20%	0.55	4,050
30% (baseline)	**0.37**	**16,500**
40%	0.37	29,850
50%	0.14	46,100

Customer value

Customer value	Optimal threshold	Max profit
300	0.63	2,730
500 (baseline)	**0.37**	**16,500**
800	0.14	42,720
1,200	0.13	83,570

Contact cost

Contact cost	Optimal threshold	Max profit
20	0.13	35,480
50 (baseline)	**0.37**	**16,500**
80	0.63	5,160
120	0.78	750



Figure 6. Maximum expected profit for combinations of customer value and campaign success rate (contact cost = 50).

Findings from the sensitivity analysis:

- **The optimal threshold is not a universal constant.** It moves between about 0.13 and 0.78 depending on assumptions, but it is below 0.50 in most scenarios, so using the default 0.50 is rarely optimal.
- **Campaign success rate is the most critical assumption.** At 15% success profit nearly vanishes (1,150), and at 50% it is about 40 times higher. It should be measured with a pilot A/B test.
- **Cheaper channels are as valuable as better models.** Reducing contact cost from 50 to 20 more than doubles maximum profit, so low-cost channels such as SMS or e-mail can matter as much as model improvements.
- **Some scenarios are not worth running.** With customer value 300 and success rate 15%, the maximum profit is zero, meaning no campaign should be launched.
- **Customer value should be individual.** At high values the threshold collapses to its floor, so a customer-specific lifetime value (CLV) would allow better targeting than one flat number.

8. Interactive Dashboard

To make the model usable by non-technical staff, it was deployed as a Streamlit web application. A user enters a customer profile (demographics, services, contract, payment method, charges) and presses "Predict". The app then returns:

- The churn probability for that customer.
- A risk flag (HIGH RISK or LOW RISK) based on an adjustable decision threshold, defaulting to the profit-optimal 0.37.
- A bar chart of the top eight SHAP contributions explaining that specific prediction.
- Rule-based retention suggestions for high-risk customers.

For a high-risk test customer (new, month-to-month, fiber optic, electronic check, no tech support), the dashboard returned a HIGH RISK status and recommended four actions: offer a discount to move to a one- or two-year contract; offer free or discounted Tech Support and Online Security; encourage automatic payment; and add an onboarding or loyalty offer for new customers. Each of these maps directly to a finding in the EDA and SHAP analyses.

Implementation notes: the trained model and its column list were saved with joblib; the app was run inside Google Colab behind a public tunnel for demonstration. For permanent hosting, the same files can be deployed on Streamlit Community Cloud.

9. Business Recommendations

Segment / driver	Evidence	Recommended action
Month-to-month customers	About 42% churn vs about 3% on two-year contracts	Offer incentives to switch to one- or two-year contracts
New customers (first months)	Churn concentrated in early tenure; tenure is the top SHAP feature	Onboarding programme, early check-in, loyalty offer
Fiber optic customers	About 42% churn; high SHAP impact	Review pricing and service quality; bundle Tech Support and Online Security
Electronic check payers	About 45% churn	Encourage automatic card or bank payment with a small incentive
Senior citizens	Higher risk in SHAP	Simpler support channels, phone assistance
Campaign design	Profit highly sensitive to cost and success rate	Use low-cost channels first; run an A/B test before scaling

10. Limitations

- **Assumed economics.** Customer value, contact cost and success rate are illustrative. Real values must come from company data and a pilot experiment.
- **Optimistic profit estimate.** The threshold was chosen on the same test set on which profit was reported. In production it should be selected on a separate validation set and confirmed on new data.
- **Correlation, not causation.** SHAP and EDA describe associations. Whether an intervention actually reduces churn must be tested experimentally.

- **Single, static dataset.** The data is a snapshot from one company; it has no time dimension, so drift over time and seasonality cannot be assessed.
- **Probability calibration.** Model probabilities were not formally calibrated, so thresholds based on cost ratios are approximate.
- **Uniform customer value.** Using one value for all customers ignores differences in lifetime value.
- **Limited tuning.** Hyperparameters were set by hand rather than through a full search; the near-identical results of three models suggest the dataset, not the algorithm, limits performance.

11. Conclusion and Future Work

This project delivered a complete analytics pipeline: data cleaning, exploratory analysis, comparison of three models, SHAP explanations, a profit-based decision rule with sensitivity analysis, and a deployable dashboard. The central lessons are that (1) a well-chosen decision threshold improved recall on churners from 0.54 to about 0.71 without changing the algorithm, (2) the main churn drivers are short tenure, flexible contracts, fiber optic service and electronic check payment, and (3) the profitability of any retention campaign hinges on the campaign's real success rate and cost, which should be measured before scaling.

Possible extensions:

- Estimate customer lifetime value (CLV) and use customer-level economics in the profit function.
- Calibrate predicted probabilities and select thresholds on a proper validation set.
- Add uplift modeling to target customers whose behavior is actually changed by a campaign, instead of simply those most likely to churn.
- Tune hyperparameters systematically and test additional models such as LightGBM.
- Add monitoring for data drift and automated retraining in production.

Appendix: Technical Summary

Component	Details
Language and libraries	Python; pandas, NumPy, scikit-learn, XGBoost, SHAP, Matplotlib, Seaborn, joblib, Streamlit
Data source	Kaggle: blastchar/telco-customer-churn (IBM sample data)
Split	80% train / 20% test, stratified, random_state = 42
XGBoost parameters	n_estimators = 300, learning_rate = 0.05, max_depth = 4, eval_metric = logloss
Random Forest parameters	n_estimators = 300, max_depth = 8, min_samples_leaf = 5
Logistic Regression	StandardScaler + LogisticRegression (max_iter = 1000)
Cross-validation	Stratified 5-fold on training set, scoring = ROC AUC
Final threshold	0.37 (profit-optimal under baseline assumptions)
Explainability	SHAP TreeExplainer on the XGBoost model